

**SA****BASIC USER MANUAL**

# Session Analytics Dashboard

User guide for the implemented admin analytics frontend, routes, forms, controls, and verified local API behavior.

**SERVICE NAME****Session Analytics Dashboard****MODULE PATH****frontend/session-analytics-  
dashboard****VERIFIED URL****<http://localhost:3002>****VERIFIED DATE****July 6, 2026**

Prepared as a simple, screenshot-based operational manual. No fake features, fake accounts, or fake workflows are included.

DOCUMENT CONTROL

# Table of Contents

|                                                |         |
|------------------------------------------------|---------|
| 1. Service Overview                            | Page 3  |
| 2. Local Access and Dependencies               | Page 4  |
| 3. New User Step-by-Step Guide                 | Page 6  |
| 4. Existing/Test User Guide                    | Page 8  |
| 5. Navigation and Common Controls              | Page 9  |
| 6. Page Guides with Screenshots                | Page 11 |
| 7. Buttons, Fields, Messages, and User Actions | Page 22 |
| 8. Common Errors and Troubleshooting           | Page 27 |
| 9. Limitations and Future Improvements         | Page 29 |

## Verification Summary

| ITEM                   | VERIFIED RESULT                                                                                                     |
|------------------------|---------------------------------------------------------------------------------------------------------------------|
| Frontend health        | <code>http://localhost:3002/healthz</code> returned <code>200</code> .                                              |
| API Gateway health     | <code>/health/live</code> and <code>/health/ready</code> returned <code>200</code> .                                |
| Session Service health | <code>http://localhost:8086/healthz</code> returned <code>200</code> .                                              |
| Seeded admin login     | Verified with the local operations admin account from the repo runbook.                                             |
| Analytics data state   | Current local stack had no active sessions, heatmap points, retention data, report schedules, or deletion requests. |

The screenshots are HD captures from the running local dashboard. They show real current local behavior, including empty states and the current funnel error state.

## CHAPTER 1

## Service Overview

Session Analytics Dashboard is an admin-facing frontend used to inspect ecommerce session activity. It is not the buyer storefront. It does not provide customer registration, product browsing, wishlist, cart, checkout, payment, profile, or notification workflows.

The implemented dashboard focuses on operational analytics:

- Overview metrics for current session activity.
- Live active sessions, device/location breakdowns, and journey links.
- Session journey lookup by session ID.
- Product-to-paid funnel analytics.
- Click and scroll heatmap aggregates.
- New, returning, and cohort retention analytics.
- CSV report export and report scheduling.
- Privacy masking, retention, and deletion request controls.

## Feature Status

| FEATURE                | STATUS    | NOTES                                                                                   |
|------------------------|-----------|-----------------------------------------------------------------------------------------|
| Admin login and logout | Available | Uses <code>/api/v1/auth/login</code> and browser <code>sessionStorage</code> .          |
| Overview               | Available | Verified. Current local data returned no matching sessions.                             |
| Live sessions          | Available | Verified. Current local data returned zero active sessions.                             |
| Journey explorer       | Available | Requires a valid session ID or a live-session row.                                      |
| Funnels                | Partial   | UI and API integration exist. Local verification displayed "Funnel report unavailable". |
| Heatmaps               | Available | Verified with an empty aggregate state.                                                 |
| Cohorts                | Available | Verified with an empty aggregate state.                                                 |
| Reports                | Available | CSV export and schedules are implemented. No schedules existed in local data.           |

|                                                        |                 |                                                                                   |
|--------------------------------------------------------|-----------------|-----------------------------------------------------------------------------------|
| Privacy controls                                       | Available       | Masking and deletion flows are available. Retention update is restricted by role. |
| Registration, product listing, cart, checkout, payment | Not Implemented | Not present in this frontend/module.                                              |

## CHAPTER 2

## Local Access and Dependencies

### Recommended Docker Run

The local Docker app is exposed on port `3002`. The root compose file builds the dashboard from `frontend/session-analytics-dashboard/Dockerfile`.

```
docker compose up -d --build session-analytics-dashboard
```

For the full platform, start all dependencies:

```
docker compose up -d --build
```

### Manual Frontend Run

The frontend is a React, TypeScript, and Vite app. The runbook notes that the Vite default port is `5174`, and recommends a different port if another dashboard is already using it.

```
cd frontend/session-analytics-dashboard
npm install
npm run dev -- --port 5175
```

### Environment Variables

| VARIABLE                             | PURPOSE                                                         | DEFAULT/EXAMPLE                    |
|--------------------------------------|-----------------------------------------------------------------|------------------------------------|
| <code>VITE_API_BASE_URL</code>       | Optional API base URL. Empty value uses the current app origin. | Blank                              |
| <code>VITE_API_PROXY_TARGET</code>   | Vite proxy target for local API calls.                          | <code>http://localhost:8080</code> |
| <code>VITE_REQUEST_TIMEOUT_MS</code> | Frontend request timeout used by the HTTP helper.               | <code>10000</code>                 |

### Required Backend/API Dependencies

| DEPENDENCY  | LOCAL URL                          | PURPOSE                              |
|-------------|------------------------------------|--------------------------------------|
| API Gateway | <code>http://localhost:8080</code> | Routes auth and analytics API calls. |

|                          |                                            |                                                                                    |
|--------------------------|--------------------------------------------|------------------------------------------------------------------------------------|
| Auth Service             | <code>http://localhost:8081/readyz</code>  | Validates login and issues admin access tokens.                                    |
| Session Service          | <code>http://localhost:8086/healthz</code> | Owns sessions, events, heatmaps, retention, reports, and privacy analytics APIs.   |
| MongoDB session database | <code>localhost:27017</code>               | Stores sessions, events, heatmap points, aggregates, reports, and privacy records. |
| Redis                    | <code>localhost:6379</code>                | Stores active session indexes and live counters.                                   |
| Session retention worker | Docker worker                              | Applies retention cleanup for session analytics data.                              |

## Important API Endpoints Used by the UI

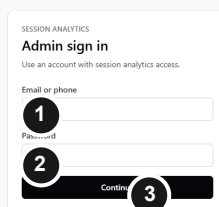
| AREA          | ENDPOINT                                                                            |
|---------------|-------------------------------------------------------------------------------------|
| Login         | <code>POST /api/v1/auth/login</code>                                                |
| Overview      | <code>GET /api/v1/analytics/live</code>                                             |
| Live sessions | <code>GET /api/v1/analytics/sessions</code>                                         |
| Journey       | <code>GET /api/v1/analytics/sessions/{sessionId}/journey</code>                     |
| Funnels       | <code>GET /api/v1/analytics/funnels</code>                                          |
| Heatmaps      | <code>GET /api/v1/analytics/heatmaps</code>                                         |
| Cohorts       | <code>GET /api/v1/analytics/retention</code>                                        |
| Reports       | <code>GET /api/v1/analytics/reports/export</code> , <code>/reports/schedules</code> |
| Privacy       | <code>/api/v1/analytics/privacy/*</code>                                            |

## CHAPTER 3

# New User Step-by-Step Guide

## Step 1: Open the Application

1. Start the local stack or the dashboard service.
2. Open `http://localhost:3002` in a browser.
3. If no admin session exists, the app redirects to `/login`.



Screenshot: Admin sign in page.

### 1 Email or phone

Enter an admin identifier.

### 2 Password

Enter an admin password with at least 8 characters.

### 3 Continue

Submits the login request.

## Step 2: Register New Account

**Not Implemented** Registration is not present in this dashboard. Admin accounts must be created by backend seed data, migrations, or an external admin/user management process.

## Step 3: Login

1. Enter a valid admin email or phone in the first field.
2. Enter the password.
3. Select **Continue**.

If either field is missing or the password is shorter than 8 characters, the screen shows:  
**Enter your admin identifier and password.**



## Step 4: Browse the Dashboard

After login, the app opens the Overview page. Use the left sidebar to switch between Overview, Live Sessions, Journey Explorer, Funnels, Heatmaps, Cohorts, Reports, and Privacy.

## Step 5: Search and Filter

Search exists for active sessions and journey lookup. Global segment filters exist on Overview, Funnels, Cohorts, and Reports. Product search and category browsing are not implemented in this module.

## Step 6: Product, Wishlist, Cart, Checkout, Payment, Profile, Notifications

| USER FLOW                  | STATUS IN THIS MODULE | INSTRUCTION                       |
|----------------------------|-----------------------|-----------------------------------|
| Product/service listing    | Not Implemented       | No listing page exists.           |
| Product/service details    | Not Implemented       | No detail page exists.            |
| Wishlist                   | Not Implemented       | No wishlist workflow exists.      |
| Cart                       | Not Implemented       | No cart workflow exists.          |
| Checkout/order flow        | Not Implemented       | No checkout workflow exists.      |
| Payment flow               | Not Implemented       | No payment UI exists.             |
| Profile/account management | Partial               | Only admin session logout exists. |
| Notifications              | Not Implemented       | No notification center exists.    |

## Step 7: Logout

Select **Sign out** at the bottom of the sidebar. The app clears the admin session and returns to the login screen.

## CHAPTER 4

## Existing/Test User Guide

### Seeded Local Admin

The repo local access guide lists a real seeded local account for this dashboard:

| FIELD             | VALUE                                                                                              |
|-------------------|----------------------------------------------------------------------------------------------------|
| Email/Login       | <code>analytics.admin.local@example.com</code>                                                     |
| Password          | <code>LocalDemo#2026!</code>                                                                       |
| Role              | <code>operations_admin</code>                                                                      |
| Credential source | Auth, user, and superadmin migration seed files referenced by <code>LOCAL_ACCESS_GUIDE.md</code> . |

### How an Existing User Logs In

1. Open `http://localhost:3002/login`.
2. Enter `analytics.admin.local@example.com`.
3. Enter `LocalDemo#2026!`.
4. Select **Continue**.

These credentials are local/demo only. Do not use them for production or shared non-local deployments.

## CHAPTER 5

## Navigation and Common Controls

### Sidebar Menu

| MENU             | ICON MEANING   | ACTION                                             |
|------------------|----------------|----------------------------------------------------|
| Overview         | Gauge          | Shows top-level live metrics and segment filters.  |
| Live Sessions    | Activity pulse | Shows active users/sessions and live breakdowns.   |
| Journey Explorer | Route path     | Looks up one session journey by session ID.        |
| Funnels          | Filter funnel  | Shows ordered conversion steps.                    |
| Heatmaps         | Pointer/click  | Shows click or scroll heatmap aggregates.          |
| Cohorts          | Repeat arrows  | Shows retention cohort analytics.                  |
| Reports          | Download       | Exports CSV reports and manages schedules.         |
| Privacy          | Shield         | Manages masking, retention, and deletion controls. |
| Sign out         | Logout arrow   | Ends the current admin session.                    |

### Common Date Controls

| CONTROL | WHAT IT DOES                                           |
|---------|--------------------------------------------------------|
| Today   | Sets From and To to the current date.                  |
| 7D      | Sets a 7-day range ending today.                       |
| 30D     | Sets a 30-day range ending today.                      |
| From    | Manual start date. Changing it makes the range custom. |
| To      | Manual end date. Changing it makes the range custom.   |

### Common Segment Filters

| FILTER | OPTIONS | USED ON |
|--------|---------|---------|
|--------|---------|---------|

|           |                                                          |                                     |
|-----------|----------------------------------------------------------|-------------------------------------|
| Device    | All devices, Desktop, Mobile, Tablet                     | Overview, Funnels, Cohorts, Reports |
| Channel   | All channels, User web, Mobile web, Android app, iOS app | Overview, Funnels, Cohorts, Reports |
| Source    | All sources, Direct, Search, Paid, Social, Email         | Overview, Funnels, Cohorts, Reports |
| User Type | All users, Anonymous, Logged in                          | Overview, Funnels, Cohorts, Reports |

## Common States

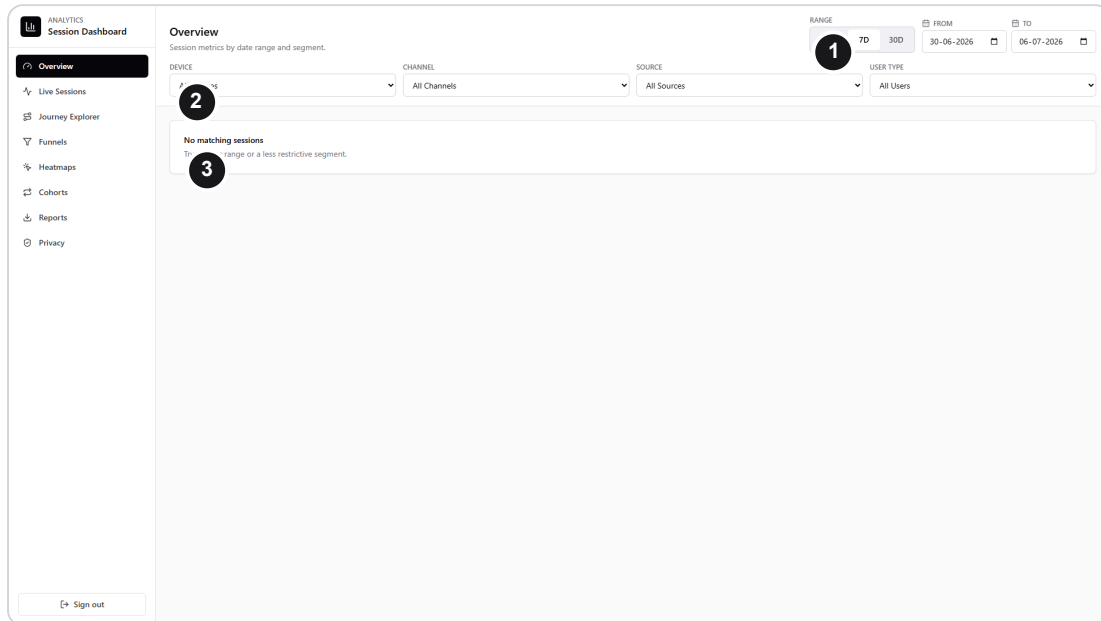
| STATE      | HOW IT APPEARS                          | USER ACTION                                                  |
|------------|-----------------------------------------|--------------------------------------------------------------|
| Loading    | Grey animated placeholder blocks.       | Wait for data to load.                                       |
| Empty      | Message such as "No matching sessions". | Widen the date range or reduce filters.                      |
| Error      | Red panel with a Retry button.          | Select Retry, then check API/backend health if it continues. |
| Validation | Amber or red inline message.            | Correct the field or date range.                             |

## CHAPTER 6

# Page Guides with Screenshots

## 6.1 Overview

The Overview page displays the current summary for selected date and segment filters. In the verified local stack, no matching sessions were available.



Screenshot: Overview page with current empty data state.

### 1 Date range

Use presets or From/To date inputs.

### 2 Segments

Filter by device, channel, source, and user type.

### 3 Empty state

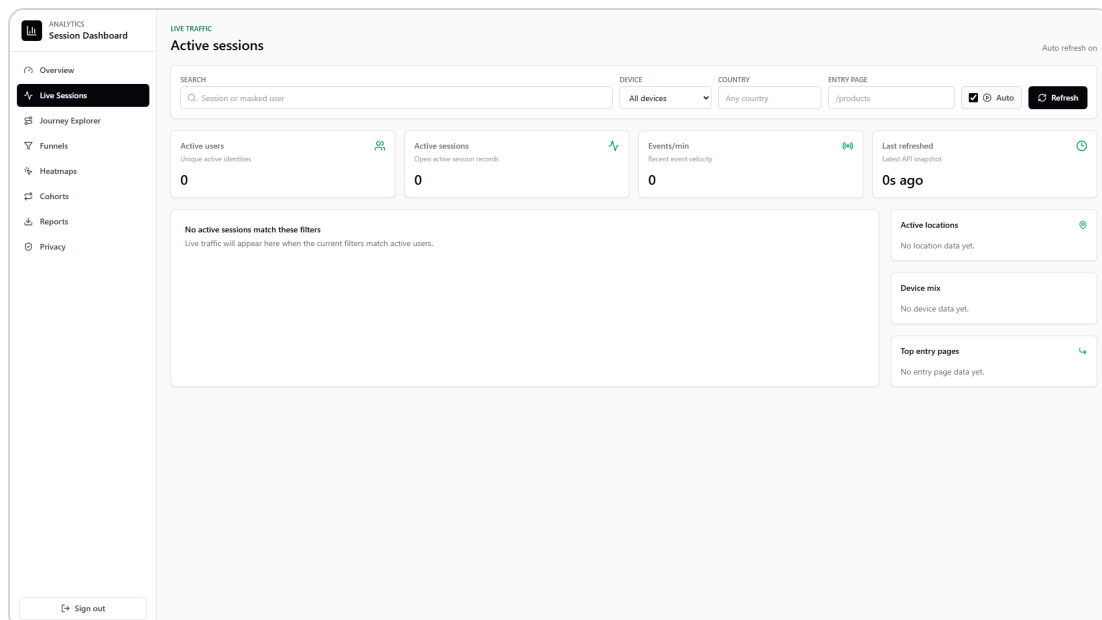
Shown when no sessions match.

## User Steps

1. Select a date preset or choose From and To dates.
2. Select segment filters.
3. Review metric cards if data exists, or follow the empty-state suggestion.

## 6.2 Live Sessions

Live Sessions shows current active traffic, active-session counts, location/device/entry-page summaries, and a live session table when data exists.

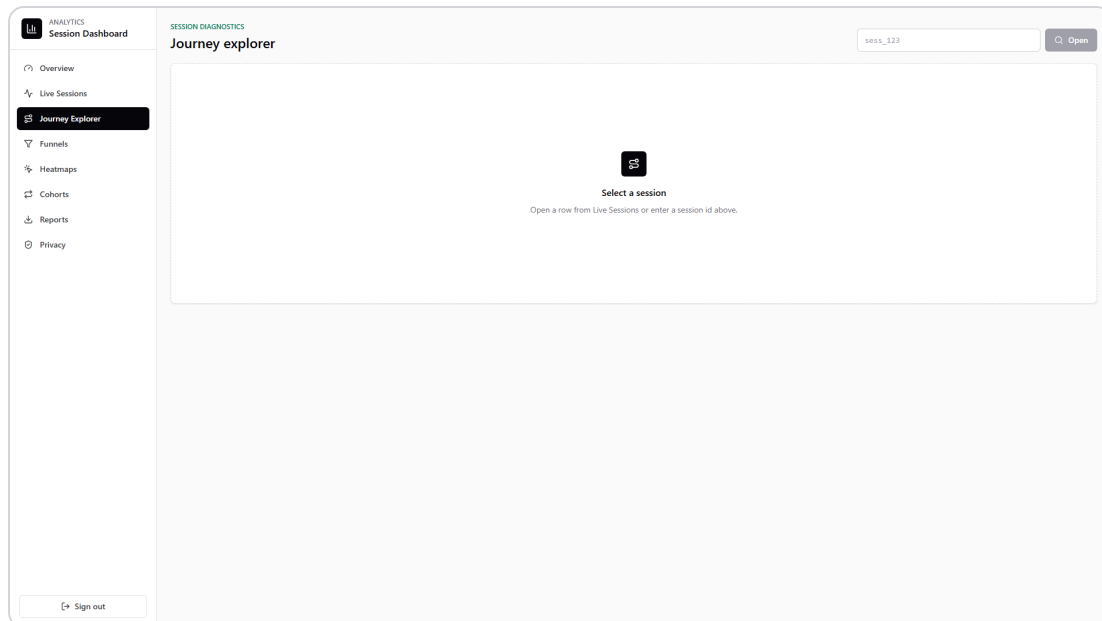


Screenshot: Live Sessions page with zero active sessions.

| CONTROL    | DESCRIPTION                                                         |
|------------|---------------------------------------------------------------------|
| Search     | Filters by session or masked user text. Max length: 128 characters. |
| Device     | Filters by all, desktop, mobile, tablet, or unknown.                |
| Country    | Optional text filter. Max length: 80 characters.                    |
| Entry page | Optional path filter. Max length: 512 characters.                   |
| Auto       | Turns automatic refresh on or off.                                  |
| Refresh    | Manually refreshes the active-session snapshot.                     |

## 6.3 Journey Explorer

Journey Explorer opens an ordered event timeline for one session. It starts with a session ID input. A user can also open it from a Live Sessions table row when live session rows exist.

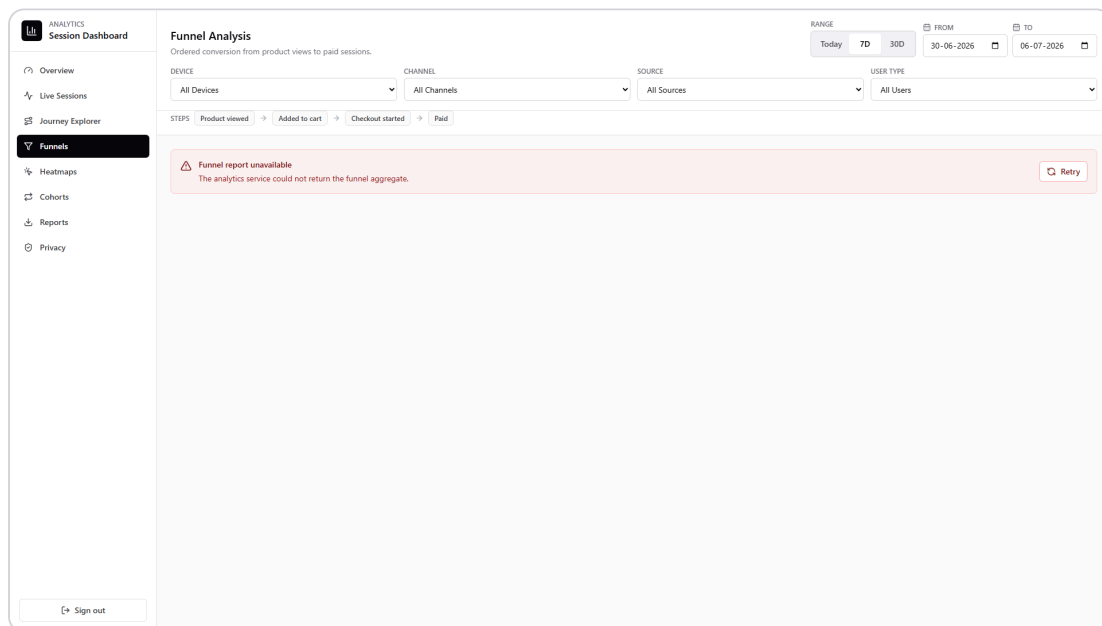


Screenshot: Journey Explorer waiting for a session ID.

1. Enter a session ID, for example `sess_123`.
2. Select **Open**.
3. If the session exists, review the session context card.
4. Use event-type filter buttons to narrow the timeline.
5. Select a timeline event to open sanitized event properties.

## 6.4 Funnel Analysis

Funnel Analysis uses fixed steps: Product viewed -> Added to cart -> Checkout started -> Paid. The UI is implemented, but during local verification the funnel API displayed an unavailable state.

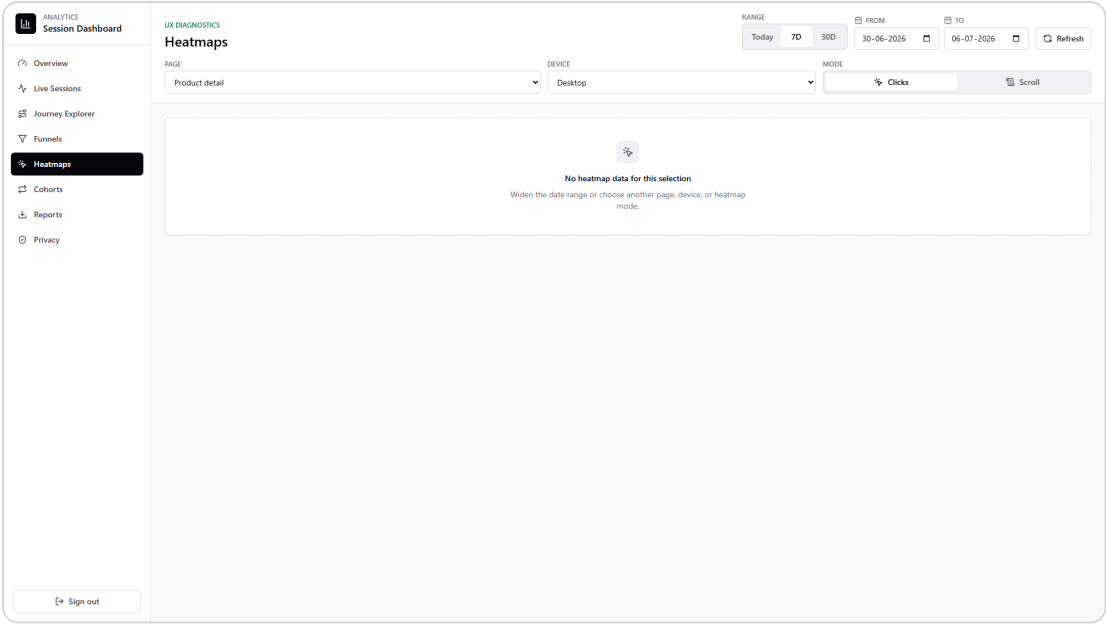


Screenshot: Funnel Analysis current error state.

Verified status: **Partial**. The screen and endpoint integration are present, but the local stack returned "Funnel report unavailable" during screenshot verification.

# 6.5 Heatmaps

Heatmaps shows click or scroll aggregates for selected pages and device classes. The verified local data had no heatmap points.



Screenshot: Heatmaps page with no data for the selected page/device/mode.

| CONTROL | DESCRIPTION                             |
|---------|-----------------------------------------|
| Page    | Selects the page path group to inspect. |
| Device  | Selects Desktop, Tablet, or Mobile.     |
| Mode    | Switches between Clicks and Scroll.     |
| Refresh | Reloads the selected heatmap aggregate. |

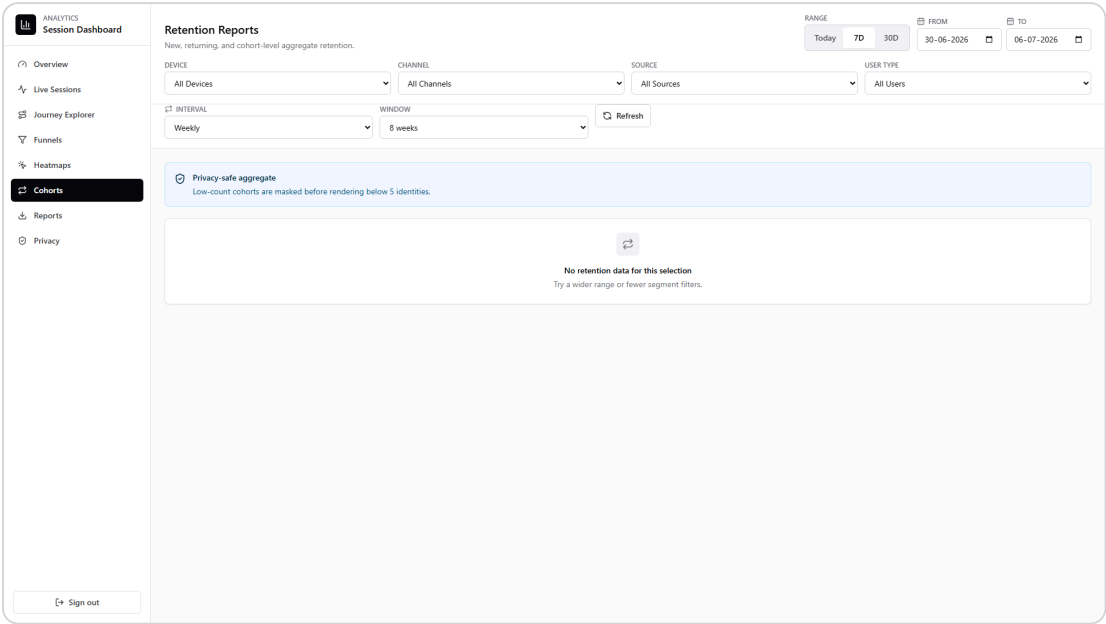


CHAPTER 6 CONTINUED

# Page Guides with Screenshots

## 6.6 Cohorts

Cohorts shows retention reports by interval and window. It includes a privacy-safe aggregate notice because low-count cohorts may be masked.

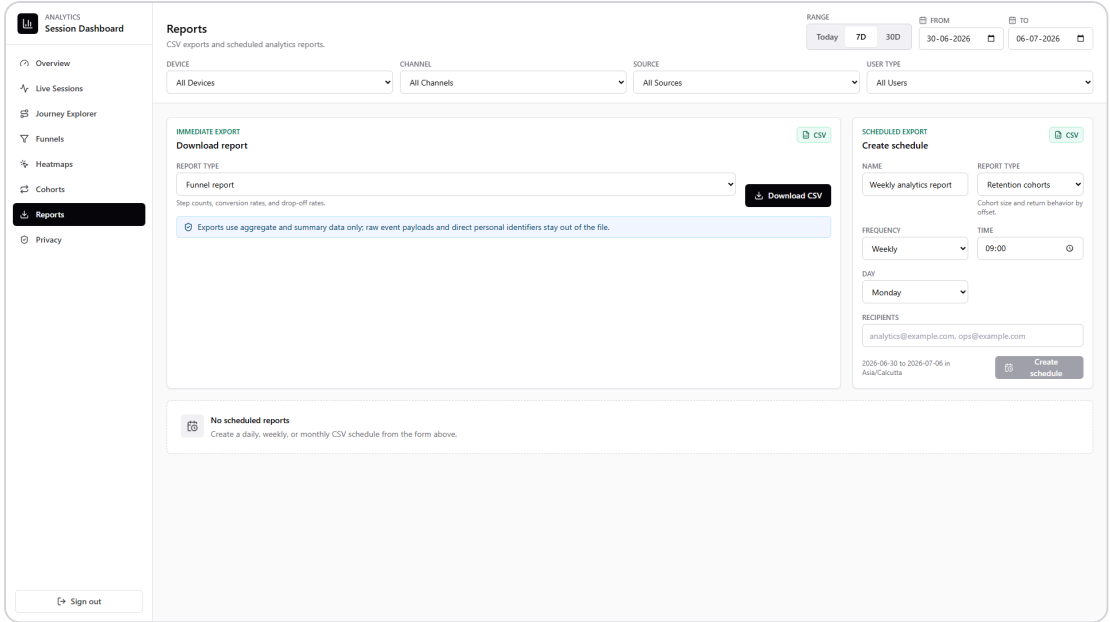


Screenshot: Cohorts page with no retention data.

| CONTROL  | DESCRIPTION                                       |
|----------|---------------------------------------------------|
| Interval | Daily, Weekly, or Monthly.                        |
| Window   | Retention window. Options depend on the interval. |
| Refresh  | Reloads the retention report.                     |

## 6.7 Reports

Reports provides immediate CSV downloads and scheduled CSV report setup. In the verified local state, there were no saved schedules.



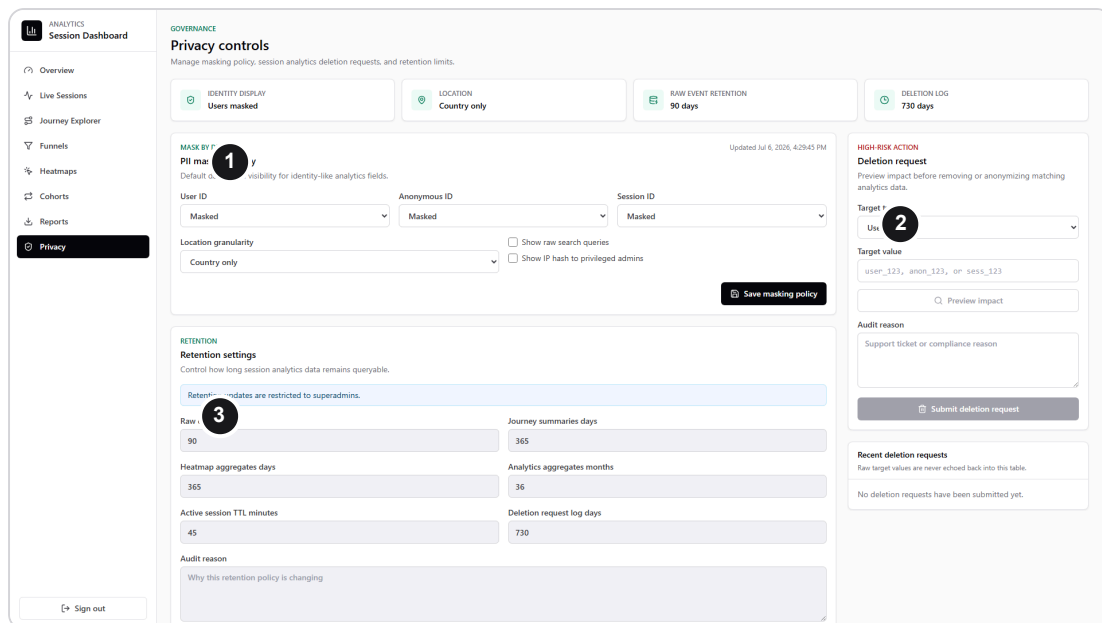
Screenshot: Reports page with download and schedule controls.

| AREA                    | FIELDS AND ACTIONS                                                                     |
|-------------------------|----------------------------------------------------------------------------------------|
| Download report         | Report Type select and <b>Download CSV</b> .                                           |
| Create schedule         | Name, Report Type, Frequency, Time, Day, Recipients, and <b>Create schedule</b> .      |
| Scheduled reports table | Shows saved schedules. When rows exist, pause/resume and delete actions are available. |

Recipients must be valid email addresses. The schedule button remains disabled until a valid name, date range, and recipient list are present.

### 6.8 Privacy Controls

Privacy controls manage masking, deletion requests, and retention values. The verified operations admin can view and update masking policy, can request deletion, and can view retention settings. Retention updates are restricted to superadmins.



**GOVERNANCE**

### Privacy controls

Manage masking policy, session analytics deletion requests, and retention limits.

**IDENTITY DISPLAY**  
Users masked

**LOCATION**  
Country only

**RAW EVENT RETENTION**  
90 days

**DELETION LOG**  
730 days

**MASK BY** **1**

PII masking visibility for identity-like analytics fields.

User ID: Masked Anonymous ID: Masked Session ID: Masked

Location granularity: Country only ☐ Show raw search queries ☐ Show IP hash to privileged admins

**Save masking policy**

**RETENTION**

**Retention settings**

Control how long session analytics data remains queryable.

Retention updates are restricted to superadmins.

**3**

Raw: 90 Journey summaries days: 365

Heatmap aggregates days: 365 Analytics aggregates months: 36

Active session TTL minutes: 45 Deletion request log days: 730

Audit reason: Why this retention policy is changing

**HIGH RISK ACTION**

**Deletion request**

Preview impact before removing or anonymizing matching analytics data.

Target value: **2**

Us: user\_123, anon\_123, or sess\_123

**Preview impact**

Audit reason: Support ticket or compliance reason

**Submit deletion request**

**Recent deletion requests**

Raw target values are never echoed back into this table.

No deletion requests have been submitted yet.

Screenshot: Privacy Controls page.

### 1 Masking policy

Controls display of user, anonymous, session, location, search, and IP hash fields.

### 2 Deletion request

Preview and submit high-risk deletion requests.

### 3 Retention settings

View or update retention rules depending on role.

## CHAPTER 7

# Buttons, Fields, Messages, and User Actions

## Authentication

| ELEMENT        | TYPE           | USER ACTION                    | VALIDATION/MESSAGE                   |
|----------------|----------------|--------------------------------|--------------------------------------|
| Email or phone | Text input     | Enter admin identifier.        | Required.                            |
| Password       | Password input | Enter admin password.          | Must be at least 8 characters.       |
| Continue       | Button         | Submit login.                  | Shows "Signing in..." while pending. |
| Sign out       | Button         | Clear session and go to login. | No confirmation dialog.              |

## Analytics Filters

| ELEMENT                            | TYPE              | USER ACTION                | VALIDATION/MESSAGE                                                                   |
|------------------------------------|-------------------|----------------------------|--------------------------------------------------------------------------------------|
| Today, 7D, 30D                     | Segmented buttons | Set a preset date range.   | Selected preset is highlighted.                                                      |
| From / To                          | Date inputs       | Set custom dates.          | "Use a valid start and end date."; "Start date must be before or equal to end date." |
| Device, Channel, Source, User Type | Selects           | Filter analytics requests. | Invalid request values are rejected before fetch by API client validation.           |

## Live Sessions

| ELEMENT | TYPE       | USER ACTION                       | MESSAGE/STATE                          |
|---------|------------|-----------------------------------|----------------------------------------|
| Search  | Text input | Filter by session or masked user. | Max 128 characters.                    |
| Device  | Select     | Choose device segment.            | All, desktop, mobile, tablet, unknown. |

|            |            |                                            |                                                          |
|------------|------------|--------------------------------------------|----------------------------------------------------------|
| Country    | Text input | Filter by country.                         | Max 80 characters.                                       |
| Entry page | Text input | Filter by entry path.                      | Max 512 characters.                                      |
| Auto       | Checkbox   | Toggle automatic refresh.                  | Header shows "Auto refresh on" or "Auto refresh paused". |
| Refresh    | Button     | Manually reload snapshot.                  | Icon spins while refreshing.                             |
| Journey    | Row link   | Open selected session in Journey Explorer. | Available only when session rows exist.                  |

## Journey Explorer

| ELEMENT            | TYPE           | USER ACTION               | MESSAGE/STATE                                                             |
|--------------------|----------------|---------------------------|---------------------------------------------------------------------------|
| Session ID         | Text input     | Enter session ID.         | Open button disabled when blank. Max 160 characters.                      |
| Open               | Button         | Loads journey route.      | 404 shows "Session not found". Other failures show "Journey unavailable". |
| Event type buttons | Toggle buttons | Filter visible events.    | Clear button appears when filters are selected.                           |
| Timeline event     | Button/card    | Select event for details. | Selected card is highlighted.                                             |
| Close event detail | Icon button    | Close detail panel.       | Returns to "Select a timeline event..." helper text.                      |

## Reports

| ELEMENT      | TYPE       | USER ACTION                | VALIDATION/MESSAGE                                                                           |
|--------------|------------|----------------------------|----------------------------------------------------------------------------------------------|
| Report Type  | Select     | Choose report type.        | Overview, Active sessions, Journey summaries, Funnel, Heatmap, Retention.                    |
| Download CSV | Button     | Downloads selected report. | Shows "Preparing CSV". Error: "Report export failed. Adjust the range or filters and retry." |
| Name         | Text input | Enter schedule name.       | Required, max 120 characters.                                                                |
| Frequency    | Select     | Daily, weekly, monthly.    | Weekly shows day-of-week; monthly shows day-of-month.                                        |

|                 |            |                                       |                                                                                                |
|-----------------|------------|---------------------------------------|------------------------------------------------------------------------------------------------|
| Time            | Time input | Set scheduled run time.               | Browser time input validation applies.                                                         |
| Recipients      | Text input | Enter comma or line separated emails. | Error: "Enter valid email addresses."                                                          |
| Create schedule | Button     | Creates CSV schedule.                 | Success: "Schedule saved." Error: "Schedule could not be saved. Check recipients and cadence." |

## CHAPTER 7 CONTINUED

# Buttons, Fields, Messages, and User Actions

## Privacy Controls

| ELEMENT                           | TYPE          | USER ACTION                                   | VALIDATION/MESSAGE                                                                       |
|-----------------------------------|---------------|-----------------------------------------------|------------------------------------------------------------------------------------------|
| User ID, Anonymous ID, Session ID | Selects       | Set Hidden, Masked, or Full display.          | Full identifiers show an amber caution notice.                                           |
| Location granularity              | Select        | Set Hidden, Country only, or City.            | Policy-controlled display.                                                               |
| Show raw search queries           | Checkbox      | Allow raw query display.                      | Disabled when permission is missing.                                                     |
| Show IP hash to privileged admins | Checkbox      | Allow IP hash display.                        | Shows caution notice when enabled.                                                       |
| Save masking policy               | Button        | Updates masking settings.                     | Error: "Masking policy could not be saved. Check permissions and retry."                 |
| Retention numeric fields          | Number inputs | Update retention values.                      | Operations admin sees read-only values. Superadmin required for updates.                 |
| Audit reason                      | Textarea      | Explain high-risk retention/deletion changes. | Must be 10 to 512 characters.                                                            |
| Save retention settings           | Button        | Saves retention settings.                     | Disabled unless role, reason, and ranges are valid.                                      |
| Target type                       | Select        | User ID, Anonymous ID, or Session ID.         | Controls validation label.                                                               |
| Target value                      | Text input    | Enter target identifier.                      | 3 to 160 characters, only letters, numbers, dot, underscore, colon, at sign, and hyphen. |
| Preview impact                    | Button        | Counts matching analytics records             | Error: "Deletion impact could not be previewed."                                         |

|                         |               |                                                    |                                                   |
|-------------------------|---------------|----------------------------------------------------|---------------------------------------------------|
|                         |               | before deletion.                                   |                                                   |
| Submit deletion request | Button        | Opens confirmation modal after preview and reason. | Error: "Deletion request could not be submitted." |
| Cancel / Submit request | Modal buttons | Cancel or confirm high-risk deletion.              | Confirm shows "Submitting" while pending.         |

## Retention Validation Ranges

| FIELD                       | ALLOWED RANGE     | ERROR MESSAGE PATTERN  |
|-----------------------------|-------------------|------------------------|
| Raw events days             | 7 to 180 days     | Use 7 to 180 days.     |
| Journey summaries days      | 30 to 730 days    | Use 30 to 730 days.    |
| Heatmap aggregates days     | 30 to 730 days    | Use 30 to 730 days.    |
| Analytics aggregates months | 12 to 84 months   | Use 12 to 84 months.   |
| Active session TTL minutes  | 15 to 180 minutes | Use 15 to 180 minutes. |
| Deletion request log days   | 365 to 2555 days  | Use 365 to 2555 days.  |

## Heatmaps and Cohorts

| ELEMENT        | TYPE              | USER ACTION              | MESSAGE/STATE                                                                                                                           |
|----------------|-------------------|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| Heatmap Page   | Select            | Choose page path group.  | Request path must be a site path beginning with  . |
| Heatmap Device | Select            | Desktop, Tablet, Mobile. | Invalid device values are rejected by API client validation.                                                                            |
| Mode           | Segmented buttons | Clicks or Scroll.        | Selected mode is highlighted.                                                                                                           |
| Interval       | Select            | Daily, Weekly, Monthly.  | Changing interval resets the window if needed.                                                                                          |
| Window         | Select            | Choose retention window. | Allowed windows depend on interval.                                                                                                     |
| Refresh        | Button            | Reloads aggregate.       | Shows "Refreshing" while pending.                                                                                                       |



## CHAPTER 8

# Common Errors and Troubleshooting

| ERROR/STATE                                            | LIKELY CAUSE                                                                            | WHAT TO DO                                                                                                                                                                             |
|--------------------------------------------------------|-----------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Login says "Enter your admin identifier and password." | Identifier is blank or password is shorter than 8 characters.                           | Enter both fields and retry.                                                                                                                                                           |
| Login fails or returns unauthorized.                   | Wrong credentials, missing seed migrations, auth service unavailable, or gateway issue. | Verify <code>http://localhost:8080/health/ready</code> and <code>http://localhost:8081/readyz</code> . Re-run <code>auth/user/superadmin</code> migrations if needed.                  |
| Overview says "No matching sessions".                  | No recent session data exists for selected range/segments.                              | Generate traffic in the user app or widen filters/date range.                                                                                                                          |
| Live sessions show zero active sessions.               | Redis has no current active session counters.                                           | Generate live traffic and keep sessions active, then refresh.                                                                                                                          |
| Funnel report unavailable.                             | Funnel aggregate endpoint returned an error in the local stack.                         | Check gateway/session-service logs and confirm funnel events use <code>product_view</code> , <code>add_to_cart</code> , <code>checkout_step</code> , and <code>payment_result</code> . |
| Heatmap has no data.                                   | No click/scroll points exist for page, device, date range, or mode.                     | Select another page/device/mode or ingest heatmap events.                                                                                                                              |
| Retention has no data.                                 | No cohort aggregate exists for the selected range/segment.                              | Use wider range or fewer filters after traffic has been ingested.                                                                                                                      |
| Report export failed.                                  | API error, invalid range, or unavailable aggregate.                                     | Adjust report type, date range, or filters and retry.                                                                                                                                  |

|                                  |                                                                               |                                                                   |
|----------------------------------|-------------------------------------------------------------------------------|-------------------------------------------------------------------|
| Retention settings are disabled. | Logged-in admin is not a superadmin.                                          | Use a superadmin account for retention updates.                   |
| Privacy/deletion action fails.   | Missing permission, invalid target value, missing reason, or backend failure. | Check role, target format, audit reason length, and service logs. |

## Useful Health Checks

```
curl http://localhost:3002/healthz
curl http://localhost:8080/health/live
curl http://localhost:8080/health/ready
curl http://localhost:8086/healthz
```

## Useful Logs

```
docker compose logs -f session-analytics-dashboard
docker compose logs -f api-gateway
docker compose logs -f session-service
docker compose logs -f auth-service
```

**CHAPTER 9**

# Final Notes, Limitations, and Future Improvements

---

## Verified Limitations

- No new-user registration exists in this dashboard.
- No buyer storefront workflows exist here: product listing, product detail, wishlist, cart, checkout, payment, profile management, and notifications are not implemented.
- The local stack had no active session data, report schedules, deletion requests, heatmap points, or retention cohorts during verification.
- Funnel Analysis displayed an unavailable state during verification.
- Retention updates are visible but disabled for the verified operations admin role.
- Journey Explorer requires a real session ID; no seeded journey ID was found in the dashboard docs or code.

## Future Improvements

- Add seeded analytics sessions/events for local demo screenshots and training.
- Expose clearer backend error details for funnel aggregate failures.
- Add a read-only profile page showing the current admin role and permissions.
- Add inline help for privacy roles and retention permissions.
- Add dashboard-level notification banners for backend outages.
- Add printable CSV/report examples once backend data is seeded.

## Manual Coverage Summary

This manual covers all implemented frontend routes, controls, forms, buttons, validation messages, navigation items, API dependencies, local access steps, and verified runtime states for the Session Analytics Dashboard. Features not present in this module are explicitly marked as not implemented.

Final deliverable: **[Session\\_Analytics\\_Dashboard\\_Basic\\_User\\_Manual.pdf](#)**